Math - 2

*Given that,

$$y = A\cos(ax-b) \quad \text{——} \ (1)$$

Now, differential w.r.to x, we get,

$$\frac{dy}{dx} = A\{-\sin(ax-b)\}.a$$

$$= -Aa\sin(ax-b)$$

Again, diff. w.r.to x, we get,

$$\frac{d^2y}{dx^2} = -Aa\cos(ax-b).a$$

$$= -Aa^2\cos(ax-b)$$

$$= -a^2A\cos(ax-b)$$

$$= -a^2y \ [\text{by eq } 1].$$

$\therefore \frac{d^2y}{dx^2} + a^2y = 0$ which required equation

Ans

*Given that,

$$y = A\cos px + B\sin px \quad \text{——} \ (1)$$

Now, diff w.r.to x, we get.

$$\frac{dy}{dx} = A(-\sin px)p + B\cos px . p$$

$= -Ap\sin px + Bp\cos px$

Again, diff w.r.to. x we get,

$$\frac{d^2y}{dx^2} = -Ap(\cos px)\cdot p + Bp(-\sin px)\cdot p$$

$$= -Ap^2\cos px - Bp^2\sin px$$

$$= -p^2(A\cos px + B\sin px)$$

$$= -p^2y \quad [\text{by eq } 1]$$

$\therefore \frac{d^2y}{dx^2} + p^2y = 0$ which is the required solution.

* variable separation:–

Given that,

$$\frac{dy}{dx} = e^{x-y} + x^2 \cdot e^{-y}$$

$$\frac{dy}{dx} = \frac{e^x}{e^y} + \frac{x^2}{e^y}$$

$$\frac{dy}{dx} = \frac{e^x + x^2}{e^y}$$

$$(e^x + x^2)\,dx = e^y\,dy.$$

Now Integrating both sides,

$$\int (e^x + x^2)\, dx = \int e^y dy$$

$$\Rightarrow e^x + \frac{x^3}{3} = e^y + c$$

which required solution. Ans

* Reduce separable variable.

Given that,

$$\frac{dy}{dx} = (4x+y+1)^2 \qquad \text{--- (1)}$$

let, $4x+y+1 = v \Rightarrow 4 + \frac{dy}{dx} = \frac{dv}{dx}$

$$\Rightarrow \frac{dy}{dx} = \frac{dv}{dx} - 4$$

Now eq^n (1) becomes,

$$\Rightarrow \frac{dv}{dx} - 4 = v^2 \Rightarrow \frac{dv}{dx} = v^2 + 2^2 \Rightarrow \frac{dv}{dx} = 2^2 + v^2$$

$$\Rightarrow \frac{dv}{2^2+v^2} = dx$$

Now Integrating both sides,

$$\int \frac{1}{2^2+v^2} dv = \int dx$$

$$\Rightarrow \frac{1}{2} \tan^{-1} \frac{v}{2} = x + c$$

$$\Rightarrow \frac{1}{2} \tan^{-1}\left(\frac{4x+y+1}{2}\right) = x + c$$

which is required solution.

Ans

*Homogeneous Differential equation:-

Given that,

$$y^2 + x^2\frac{dy}{dx} = xy\frac{dy}{dx}$$

$$\Rightarrow (x^2 - xy)\frac{dy}{dx} = -y^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{-y^2}{(x^2-xy)} \quad \text{———} \; (1)$$

let,

$$y = vx$$

$$\therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$$

So eq^n (1) can be written as,

$$v + x\frac{dv}{dx} = -\frac{v^2x^2}{x^2 - x\cdot xv}$$

$$\Rightarrow v + x\frac{dv}{dx} = -\frac{x^2(v^2)}{x^2(1-v)}$$

$$\Rightarrow v + x\frac{dv}{dx} = -\frac{v^2}{1-v}$$

$$\Rightarrow x\frac{dv}{dx} = -\frac{v^2}{1-v} - v$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-v^2 - v + v^2}{1-v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{-v}{1-v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v}{v-1}$$

$$\Rightarrow \frac{1}{x}\frac{dv}{dx} = \frac{v-1}{v}$$

$$\Rightarrow \frac{1}{x}dx = \left(\frac{v-1}{v}\right)dv$$

Now Integrating,

$$\int \frac{1}{x}dx = \int\left(\frac{v-1}{v}\right)dv$$

$$\Rightarrow \ln x + c = \int\left(1-\frac{1}{v}\right)dv$$

$$\Rightarrow \ln x + c = v - \ln v$$

$$\Rightarrow \ln x + \ln v + c = v$$

$$\Rightarrow \ln x + \ln\frac{y}{x} + c = \frac{y}{x} \quad \left[\because y = vx \therefore v = \frac{y}{x}\right]$$

$$\Rightarrow \ln\left(x\cdot\frac{y}{x}\right) + c = \frac{y}{x}$$

$$\Rightarrow \ln y + c = \frac{y}{x}$$

which is required solution Ans

*Given that,

$$\left(x\cos\frac{y}{x}+y\sin\frac{y}{x}\right)y=\left(y\sin\frac{y}{x}-x\cos\frac{y}{x}\right)x\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}=\frac{y\left(x\cos\frac{y}{x}+y\sin\frac{y}{x}\right)}{x\left(y\sin\frac{y}{x}-x\cos\frac{y}{x}\right)} \quad \text{——— (1)}$$

let,

$y=vx$

$$\therefore \frac{dy}{dx}=v+x\frac{dv}{dx}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{dy}{dx}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{y\left(x\cos\frac{y}{x}+y\sin\frac{y}{x}\right)}{x\left(y\sin\frac{y}{x}-x\cos\frac{y}{x}\right)}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{vx\left(x\cos\frac{vx}{x}+vx\sin\frac{vx}{x}\right)}{x\left(vx\sin\frac{vx}{x}-x\cos\frac{vx}{x}\right)}$$

$$\Rightarrow v+x\frac{dv}{dx}=\frac{v\left(x\cos v+vx\sin v\right)}{\left(vx\sin v-x\cos v\right)}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{v\left(x\cos v+vx\sin v\right)}{\left(vx\sin v-x\cos v\right)}-v$$

$$\Rightarrow x\frac{dv}{dx}=\frac{v\left(\cos v+v\sin v\right)}{\left(v\sin v-\cos v\right)}-v$$

$$\Rightarrow x\frac{dv}{dx}=\frac{v\left(\cos v+v\sin v\right)-v\left(v\sin v-\cos v\right)}{v\sin v-\cos v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v\cos v + v^2\sin v - v^2\sin v + v\cos v}{v\sin v - \cos v}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{2v\cos v}{v\sin v - \cos v}$$

$$\Rightarrow \frac{1}{x}\frac{dv}{dx} = \frac{v\sin v - \cos v}{2v\cos v}$$

$$\Rightarrow \frac{2}{x}dx = \left(\frac{v\sin v - \cos v}{v\cos v}\right)dv$$

$$\Rightarrow \frac{2}{x}dx = \left(\tan v - \frac{1}{v}\right)dv$$

~~Now Integrating~~,

$$\Rightarrow \frac{2}{x}dx = \left(\frac{\tan v\cdot\sec v}{\sec v} - \frac{1}{v}\right)dv \;[$$

Now Integrating;

$$\int \frac{2}{x}dx = \int\left(\frac{\tan v\cdot\sec v}{\sec v} - \frac{1}{v}\right)dv$$

or, $2\ln x + \ln c = \ln\sec v - \ln v.$

or, $2\ln x + \ln c = \ln\frac{\sec v}{v}.$

or, $\ln x^2 + \ln c = \ln\frac{\sec\frac{y}{x}}{\frac{y}{x}}.$

or, $\ln(x^2 c) = \ln\frac{\sec\frac{y}{x}}{\frac{y}{x}}$

or, $x^2 c = \dfrac{\sec\frac{y}{x}}{\frac{y}{x}}$

or, $x^2 c \cdot \frac{y}{x} = \sec\frac{y}{x}$

or, $cxy = \sec\frac{y}{x}$.

$\therefore \sec\frac{y}{x} = cxy$ which is required Solution. Ans

* Reducible Homogeneous Differential Equation

* Given that,

$(2x+y+3)\frac{dy}{dx} = x+2y+3$

or, $\frac{dy}{dx} = \frac{x+2y+3}{2x+y+3}$ —① $\left[\frac{a_1}{b_1} \neq \frac{a_2}{b_2} \text{ i.e } \frac{1}{2} \neq \frac{2}{1}\right]$.

Let, $x+y=v$.

$\Rightarrow 1+\frac{dy}{dx} = \frac{dv}{dx}$

$\Rightarrow \frac{dy}{dx} = 1-\frac{dv}{dx}$

eq ① becomes,

$1-\frac{dv}{dx} = \frac{x+2y+3}{2x+y+3}$

putting $x = X+h$ and $y = Y+k$

Such that, $\frac{dy}{dx} = \frac{dY}{dX}$.

Hence the above eqn becomes,

$$\frac{dY}{dX} = \frac{X+2Y+h+2k+3}{2X+Y+2h+k+3} \quad \text{——— (2)}$$

choosing h and k such that

$$h + 2k + 3 = 0$$
$$2h + k + 3 = 0.$$

Solving the above eqn

$$\frac{h}{6-3} = \frac{k}{6-3} = \frac{1}{1-4}$$

or, $\frac{h}{3} = \frac{k}{3} = -\frac{1}{3}$.

$\therefore h = -1, \; k = -1.$

equation (2) becomes,

$$\frac{dY}{dX} = \frac{X+2Y}{2X+Y}$$

Now putting $Y = VX$ so that, $\frac{dY}{dX} = V + X\frac{dV}{dX}$

$$\therefore V + X\frac{dV}{dX} = \frac{X+2VX}{2X+VX}$$

$$V + X\frac{dV}{dX} = \frac{1+2V}{2+V}$$

or, $x\frac{dv}{dx} = \frac{1+2v}{2+v} - v$

$\Rightarrow x\frac{dv}{dx} = \frac{1+2v-2v-v^2}{2+v}$

$\Rightarrow x\frac{dv}{dx} = \frac{1-v^2}{2+v}$

$\Rightarrow \frac{1}{x} dx = \frac{2+v}{1-v^2} dv$

$\Rightarrow \frac{1}{x} dx = \frac{2+v}{(1-v)(1+v)} dv$

$\Rightarrow \frac{1}{x} dx = \left(\frac{\frac{3}{2}}{1-v} + \frac{\frac{1}{2}}{1+v}\right) dv$

or, $\ln x = -\frac{3}{2}\ln(1-v) + \frac{1}{2}\ln(1+v) - \ln c$

or, $\ln x + \ln c = \ln \frac{(1+v)^{\frac{1}{2}}}{(1-v)^{\frac{3}{2}}}$

or, $\ln cx = \ln \frac{(1+v)^{\frac{1}{2}}}{(1-v)^{\frac{3}{2}}}$

or, $c^2x^2 = \frac{1+v}{(1-v)^3}$

or, $c^2x^2 = \frac{1+\frac{y}{x}}{\left(1-\frac{y}{x}\right)^3}$

or, $c^2x^2\left(1-\frac{y}{x}\right)^3 = 1+\frac{y}{x}$

or, $c^2(X-\cancel{Y})^2 = X+Y$

or, $c^2(\cancel{x}-1-y+1)^2 = x-1+y-1$

or, $c^2(x-y)^2 = x+y-2$

which is required solution.

* Given that,

$$(2x-2y+5)\frac{dy}{dx} = x-y+3$$

$$\Rightarrow \frac{dy}{dx} = \frac{x-y+3}{2x-2y+5} \quad \text{——— (1)}$$

let, $x-y=v$

$$\Rightarrow 1-\frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow \frac{dy}{dx} = 1-\frac{dv}{dx}$$

eqn (1) becomes,

$$1-\frac{dv}{dx} = \frac{x-y+3}{2x-2y+5}$$

$$\Rightarrow 1-\frac{dv}{dx} = \frac{v+3}{2v+5}$$

$$\Rightarrow \frac{dv}{dx} = 1-\frac{v+3}{2v+5}$$

$$\Rightarrow \frac{dv}{dx} = \frac{2v+5-\cancel{2v-5v}\ v-3}{2v+5}$$

$$\Rightarrow dx = \left(\frac{2v+5}{v+2}\right) dv$$

Now Integrating,

$$\int dx = \int \left(\frac{2v+5}{v+2}\right) dv$$

$$\Rightarrow x + c = \int \left\{\frac{2(x+2)+1}{v+2}\right\} dv$$

$$\Rightarrow x + c = \int \left(2 + \frac{1}{v+2}\right) dv$$

$$\Rightarrow x + c = 2v + \ln(v+2)$$

$$\Rightarrow x + c = 2(x-y) + \ln(x-y+2)$$

$\therefore \cancel{2x - 2y + c}$

$\Rightarrow 2y - x + c = \ln(x-y+2)$ which is required solution.

Ans

* linear Differential

Given that,

$$x\frac{dy}{dx}+2y=x^2\log x$$

or, $$\frac{dy}{dx}+\frac{2}{x}y=x\log x$$

$$I.F = e^{\int \frac{2}{x}dx}$$

$$= e^{2\log x}$$

$$= x^2$$

Hence the solution is,

$$y\cdot x^2 = c+\int x^2\cdot x\log x\,dx$$

$$= c+\int x^3\log x\,dx$$

$$= c+\log x\cdot\frac{x^4}{4}-\int\frac{1}{x}\cdot\frac{x^4}{4}dx$$

$$= c+\frac{1}{4}x^4\log x-\frac{1}{16}x^4$$

$$= cx^{-2}+\frac{1}{4}x^2\left(\log x-\frac{1}{4}\right)$$

Ans

Given that,

$$y\log y\,dx + (x-\log y)\,dy = 0.$$

$$\Rightarrow \frac{y\log y\,dx}{y\log y\,dy} + \frac{(x-\log y)\,dy}{y\log y\,dy} = 0.$$

$$\Rightarrow \frac{dx}{dy} + \frac{x-\log y}{y\log y} = 0.$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y\log y} - \frac{1}{y} = 0.$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{y\log y} = \frac{1}{y} \quad \text{--- (1)}$$

Now Integrating factor. $I.F = e^{\int \frac{1}{y\log y}dy}$

$$= e^{\int \frac{\frac{1}{y}}{\log y}dy}$$

$$= e^{\ln(\log y)}$$

$$= \log y \;\; [\because e^{\ln x} = x].$$

Now multiplying (1) I.F, we get,

$$\log y\frac{dx}{dy} + \log y\cdot\frac{x}{y\log y} = \log y\cdot\frac{1}{y}.$$

$$\Rightarrow \frac{d}{dy}(\log y\cdot x) = \frac{\log y}{y}$$

Now Integrating,

$$\int \frac{dy}{dx}(\log y . x) = \int \frac{\log y}{y} dy. \quad \Big| \begin{array}{l} z = \log y \\ \frac{1}{y} dy = dz. \end{array}$$

$\Rightarrow \log y . x = \int z\,dz$

$\Rightarrow \log y . x = \frac{z^2}{2} + c.$

$\therefore x \log y = (\log y)^2/2 + c$ which is required solution.

*Bernoulli's Differential Equation

Given that,

$$\frac{dy}{dx} = x^3y^3 - xy$$

$$\text{or } \frac{dy}{dx} + xy = x^3y^3 \text{ ——— } (1).$$

dividing eq (1) by y^3

$$y^{-3}\frac{dy}{dx} + xy^{-2} = x^3 \text{ ——— } (2)$$

let, $y^{-2} = v.$

$$\Rightarrow -2y^{-3}\frac{dy}{dx} = \frac{dv}{dx}$$

$$\Rightarrow y^{-3}\cdot\frac{dy}{dx} = -\frac{1}{2}\,dv/dx$$

So eq^n ② becomes,

$$-\frac{1}{2}\frac{dv}{dx} + xv = x^3$$

$$\Rightarrow \frac{dv}{dx} - 2xv = 2x^3 \quad \text{——— ③}$$

$$\text{So I.F} = e^{\int -2x\,dx}$$

$$= e^{-2x^2/2}$$

$$= e^{-x^2}$$

multiplying eq^n ③ by e^{-x^2}

$$e^{-x^2}\frac{dv}{dx} - e^{-x^2}\cdot 2xv = e^{-x^2}\cdot 2x^3$$

$$\Rightarrow \frac{d}{dx}\left(e^{-x^2}\cdot v\right) = -2x^3\cdot e^{-x^2}$$

Now Integrating,

let,
$x^2 = z$
$2x\,dx = dz.$

$$\Rightarrow e^{-x^2}\cdot v = \int -2x^3\cdot e^{-x^2}$$

$$\Rightarrow e^{-x^2}\cdot v = \int -z e^{-z}\,dz$$

$$\Rightarrow e^{-x^2}\cdot v = -z\int e^{-z}dz - \int\left\{\frac{d}{dz}(-z)\int e^{-z}dz\right\}dz.$$

$$\Rightarrow e^{-x^2}\cdot v = z e^{-z} - \int e^{-z}\,dz$$

$$\Rightarrow e^{-x^2} \cdot v = ze^{-z} + e^{-z} + c$$

$$\Rightarrow e^{-x^2} \cdot v = e^{-z}(z+1) + c$$

$$\Rightarrow e^{-x^2} \cdot v = e^{-x^2}(x^2+1) + c.$$

$$\Rightarrow v = x^2 + 1 + c.$$

$\Rightarrow y^{-2} = x^2 + 1 + c$ which is required Solution.

* Exact differential Equation.

Given that,

$$(x - 2e^y)dy + (y + x + x\sin x)dx = 0 ..$$

$$\Rightarrow (y + x\sin x)dx + (x - 2e^y)dy = 0 \text{ ——— } ①$$

Comparing eq^n ① with $Mdx + Ndy = 0$ we get,

$$M = y + x\sin x, \quad N = x - 2e^y$$

$$\Rightarrow \frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 1.$$

if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so the given eq^n is exact

Now, Integrating M w.r.to x keeping y constant.

$$\int (y + x\cos x)dx = \int y\,dx + \int x\sin x\,dx$$

$$= yx + x\int \sin x\,dx - \int\left\{\frac{d}{dx}x \int \sin x\,dx\right\}dx$$

$$= yx + x(-\cos x) - \int(-\cos x)\,dx$$

$$= yx - x\cos x + \sin x + C.$$

Again, x free term in N is $-2e^{y}$

so integrating $-2e^{y}$ w.r.to y we get,

$$\int -2e^{y}dy = -2\int e^{y}dx = -2e^{y} + C.$$

Hence the general solution is,

$xy - x\cos x + \sin x - 2e^{y} + C$ Ans

Given that,

$$\left(1+e^{\frac{x}{y}}\right)dx + e^{\frac{x}{y}}\left(1-x/y\right)dy = 0 \longleftarrow (1)$$

comparing (1) with $Mdx + Ndy = 0$

we get,

$$M = 1+e^{x/y} \; , \quad N = e^{x/y}.(1-x/y)$$

$$\frac{\partial M}{\partial y} = e^{x/y}\left(-\frac{x}{y^2}\right), \quad \frac{\partial N}{\partial x} = e^{x/y}\frac{1}{y}\left(1-\frac{x}{y}\right) + e^{x/y}\left(-\frac{1}{y}\right)$$

$$= e^{x/y}\left(-\frac{x}{y^2}\right)$$

As $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ so the given equation is exact.

Now. Integrating M w.r.to x keeping y constant.

$$\int Mdx = \int \left(1+e^{x/y}\right)dx$$

$$= x + \frac{e^{\frac{x}{y}}}{\frac{1}{y}}$$

$$= x + ye^{x/y}$$

In N $\left(e^{\frac{x}{y}}\left(1-\frac{x}{y}\right)\right)$ There is no term Free form x.

Hence the required solution is $x+ye^{\frac{x}{y}}=c$

Ans

Group B

~~Gi~~

Variation of parameter:-

Given that,

$$\frac{d^2y}{dx^2}+n^2y=\sec nx \quad \text{--- (1)}$$

$$\text{or, } y''+y=0 \quad \text{--- (ii)}$$

let, $y=e^{mx}$ be the trail ~~of~~ solⁿ of (ii)

$$y=me^{mx}, \quad y''=m^2e^{mx}$$

So eqⁿ (2) becomes,

$$(m^2+n^2)e^{mx}=0 \quad \text{but, } e^{mx}\neq 0.$$

$\therefore m^2+n^2=0$ which is the

A.E; $\therefore m=\pm ni$

Hence the C.F $=c_1\cos nx+c_2\sin nx$

Then the general solution is given

$$y=u(x)\cos nx+v(x)\sin nx \quad \text{--- (3)}$$

Again the wronskoin is given by,

$$W = \begin{vmatrix} \cos nx & \sin nx \\ -n\sin nx & n\cos nx \end{vmatrix}$$

$$= n(\cos^2 nx + \sin^2 nx)$$

$$= n \times 1$$

$$= n.$$

Now,

$$u(x) = -\int \frac{y_2 R(x)}{W} dx$$

$$= -\frac{1}{n}\int \sin nx \sec nx\, dx$$

$$= -\frac{1}{n}\int \sin nx \cdot \frac{1}{\cos nx} dx$$

$$= -\frac{1}{n}\int \tan nx\, dx$$

$$= +\frac{1}{n^2}\log \cos nx + c_1$$

$$v(x) = \int \frac{y_2 (Rx)}{W} dx$$

$$= \frac{1}{n}\int dx$$

$$= \frac{1}{n}x + c_2$$

Now, putting the values of $u(x)$ and $v(x)$ in eq^n (3)

we get,

$$y = u(x)\cos nx + v(x)\sin nx$$

$$= \left(\frac{1}{n^2}\log \cos nx + c_1\right)\cos nx + \left(\frac{x}{n} + c_2\right)\sin nx$$

Ans

* Given that,

$$\frac{d^2y}{dx^2} + 4y = 4\tan 2x$$

$$\text{or, } (D^2+4)y = 4\tan 2x \text{ ——— (1)}$$

$$\text{or, } (D^2+4)y = 0 \text{ ——— (2)}$$

let, $y = e^{mx}$ be the trail sol^n of eq^n (2)

$m^2 + 4 = 0$ which is the A.E.

$$m = \pm 2i$$

Hence the C.F,

$$y = c_1\cos 2x + c_2\sin 2x$$

Therefore the general solution is given

$$y = u(x)\cos 2x + v(x)\sin 2x \text{ ——— (3)}$$

the wronskian is given by

$$W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$= 2(\cos^2 2x + \sin^2 2x)$$

$$= 2.$$

Now,

$$u(x) = -\int \frac{y_2 R(x)}{W}$$

$$= -\int \frac{\sin 2x \cdot 4\tan 2x}{2} dx$$

$$= -2\int \frac{\sin 2x}{\cos 2x} \cdot dx$$

$$= -2\int \frac{(1-\cos^2 2x)}{\cos 2x} dx$$

$$= -2\int (\cos 2x - \sec 2x) dx$$

$$= \sin 2x - \log(\sec 2x + \tan 2x) + c_1$$

Now,

$$v(x) = \int \frac{y_1 R(x)}{W}$$

$$= \int \frac{\cos 2x \cdot 4\tan 2x}{2} dx$$

$$= 2\int \cos 2x \cdot \frac{\sin 2x dx}{\cos 2x}$$

$= 2\int \sin 2x \, dx$

$= -2 \cdot \frac{1}{2} \cos 2x + c_2$

$= -\cos 2x + c_2$

Now, putting the values of $u(x)$ and $v(x)$ in the eq^n (3), we get,

$y = u(x) \cos 2x + v(x) \sin 2x$

$= [\sin 2x - \log(\sec 2x + \tan 2x) + c_1] \cos 2x +$

$(-\cos 2x + c_2) \sin 2x$

$= c_1 \cos 2x + c_2 \sin 2x + \cos 2x \cdot \sin 2x - \cos 2x$

$\log(\sec 2x + \tan 2x) - \sin 2x \cdot \cos 2x$

$= c_1 \cos 2x + c_2 \sin 2x - \cos 2x \log(\sec 2x + \tan 2x)$

Ans

Undetermain co-efficients:-

Given that,

$$(D^2+2)y = e^x + 2 \quad \text{—— (1)}$$

A.E is $D^2 + 2 = 0$

$\therefore D = \pm 2i$

$\therefore$ C.F is $y = c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x$

As a particular integrating we ~~get~~, write,

$$y = Ae^x + B.$$

$\therefore Dy = Ae^x, \quad D^2y = Ae^x$

putting these value in eq^n ① we get,

$$Ae^x + 2Ae^x + 2B = e^x + 2$$

or, $3Ae^x + 2B = e^x + 2$

Equating co-efficient of like terms.

$3A = 1 \qquad B = 1$

$A = \frac{1}{3}$

Hence a particular integrant of D.E is

$$y = \frac{1}{3}e^x + 1$$

$\therefore$ The general ~~sol~~^{n} ①. $y = c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x + \frac{1}{3}e^x + 1.$

*Given that,

$$(D^2 - 2D + 3)y = x^3 + \sin x \quad \text{—— ①}$$

A.E is $D^2 - 2D + 3 = 0$.

$$\therefore D = \frac{2 \pm \sqrt{4 - 12}}{2 \cdot 1} = 1 \pm \sqrt{2}i$$

CF is $y = e^x(c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x)$

As a particular integrand we take,

$y = Ax^3 + Bx^2 + Cx + D + E\sin x + F\cos x$

$\therefore Dy = 3Ax^2 + 2Bx + C + E\cos x - F\sin x$

$\therefore D^2y = 6Ax + 2B - E\sin x - F\cos x$

From ① we get,

$6Ax + 2B - E\sin x - F\cos x - 6Ax^2 - 4Bx - 2C - 2E\cos x + 2F\sin x + 3Ax^3 + 3Bx^2 + 3Cx + 3D + 2E\sin x + 3F\cos x = x^3 + \sin x$

or, $3Ax^3 + (-6A + 3B)x^2 + (6A - 4B + 3C)x + (2B - 2C + 3D) + (2E + 2F)\sin x + (-2E + 2F)\cos x = x^3 + \sin x$

Ans

go to page ⑦

7

Bessel's Equation

* Given that,

$$J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}}\cos x$$

we know that,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{n+2r}}{\lfloor r \cdot \Gamma(n+r+1)}$$

Now, putting $n = -\frac{1}{2}$ we get,

$$J_{-\frac{1}{2}}(x) = \sum_{r=0}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{-\frac{1}{2}+2r}}{\lfloor r\, \Gamma\left(-\frac{1}{2}+r+1\right)}$$

$$= \left(\frac{x}{2}\right)^{-1/2} \sum_{r=0}^{\infty} \frac{(-1)^r \left(\frac{x}{2}\right)^{2r}}{\lfloor r\, \Gamma\left(\frac{1}{2}+r\right)}$$

$$= \left(\frac{x}{2}\right)^{-1/2} \left[\frac{1}{\Gamma\frac{1}{2}} - \frac{(x/2)^2}{\lfloor 1\, \Gamma\frac{3}{2}} + \frac{\left(\frac{x}{2}\right)^4}{\lfloor 2\, \Gamma\frac{5}{2}} + \cdots \infty\right]$$

$$= \left(\frac{2}{x}\right)^{\frac{1}{2}} \left[\frac{1}{\Gamma\frac{1}{2}} - \frac{x^2}{2^2 \lfloor 1 \cdot \frac{1}{2} \cdot \Gamma\frac{1}{2}} + \frac{x^4}{2^4 \lfloor 2 \cdot \frac{3}{2} \frac{1}{2} \cdot \Gamma\frac{1}{2}} \cdots \infty\right]$$

$$= \sqrt{\frac{2}{x}} \left[\frac{1}{\sqrt{\pi}} - \frac{x^2}{2 \cdot 1 \cdot \sqrt{\pi}} + \frac{x^4}{4 \cdot 3 \cdot 2 \cdot 1 \sqrt{\pi}} \cdots \infty\right]$$

$$= \frac{\sqrt{2}}{\sqrt{x}} \cdot \frac{1}{\sqrt{\pi}} \left\{1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} \cdots \infty\right\}$$

$$= \sqrt{\frac{2}{\pi x}} \cdot \cos x \quad \underline{\underline{\text{Ans}}}$$

* Given that,

$$\sqrt{\frac{\pi x}{2}}\, J_{\frac{3}{2}}(x) = \frac{\sin x}{x} - \cos x$$

we know that,

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r \cdot \left(\frac{x}{2}\right)^{n+2r}}{\lfloor r \cdot \Gamma(n+r+1)}$$

$$\therefore J_{\frac{3}{2}}(x) = \sum_{r=0}^{\infty} \frac{(-1)^r \cdot \left(\frac{x}{2}\right)^{\frac{3}{2}+2r}}{\lfloor r \cdot \Gamma\left(\frac{3}{2}+r+1\right)}$$

$$= \sum_{r=0}^{\infty} \frac{(-1)^r \cdot \left(\frac{x}{2}\right)^{\frac{3}{2}} \left(\frac{x}{2}\right)^{2r}}{\lfloor r \cdot \Gamma\left(\frac{5}{2}+r\right)}$$

$$= \left(\frac{x}{2}\right)^{\frac{3}{2}} \left[\frac{1}{\Gamma\frac{5}{2}} - \frac{\left(\frac{x}{2}\right)^2}{\lfloor 1 \cdot \Gamma\frac{7}{2}} + \frac{\left(\frac{x}{2}\right)^4}{\lfloor 2\, \Gamma\frac{9}{2}} \cdots\cdots \infty\right]$$

$$= \left(\frac{x}{2}\right)^{\frac{3}{2}} \left[\frac{1}{\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}} - \frac{x^2}{2^2 \cdot \lfloor 1 \cdot \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{1}\sqrt{\pi}} + \frac{x^4}{2^4 \cdot \lfloor 2 \cdot \frac{7}{5} \cdot \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{1}\sqrt{\pi}} \cdots\cdots \infty\right]$$

$$= \frac{x^{3/2}}{2^{3/2}} \cdot \frac{1}{2/3 \cdot 1/2\sqrt{\pi}} \left[1 - \frac{x^2}{2.5} + \frac{x^4}{2.4.5.7} \cdots\cdots \infty\right]$$

$$= \frac{x^{3/2}\sqrt{2}}{3\sqrt{\pi}} \cdot \frac{1}{x^2}\left[x^2 - \frac{x^4}{2.5} + \frac{x^6}{2.4.5.7} - \cdots\cdots\right].$$

$$= \sqrt{\frac{2}{\pi x}}\left[\frac{x^2}{3} - \frac{x^4}{2.3.5} + \frac{x^6}{2.3.4.5.7} - \cdots \infty\right]$$

$$= \sqrt{\frac{2}{\pi x}}\cdot\left[\frac{2x^2}{1.2.3} - \frac{4x^4}{1.2.3.4.5} + \frac{6x^6}{1.2.3.4.5.6.7} - \cdots \infty\right]$$

$$= \sqrt{\frac{2}{\pi x}}\left[\frac{2x^2}{\lfloor 3} - \frac{4x^4}{\lfloor 5} + \frac{6x^6}{\lfloor 7} - \cdots \infty\right]$$

or, $$\sqrt{\frac{\pi x}{2}}\, J_{\frac{3}{2}}(x) = \left(\frac{1}{\lfloor 2} - \frac{1}{\lfloor 3}\right)x^2 - \left(\frac{1}{\lfloor 4} - \frac{1}{\lfloor 5}\right)x^4 + \left(\frac{1}{\lfloor 6} - \frac{1}{\lfloor 7}\right)x^6 + \cdots \infty$$

or, $$\sqrt{\frac{\pi x}{2}}\, J_{\frac{3}{2}}(x) = \left(\frac{x^2}{\lfloor 2} - \frac{x^4}{\lfloor 4} + \frac{x^6}{\lfloor 6} - \cdots\right) + \left(-\frac{x^2}{\lfloor 3} + \frac{x^4}{\lfloor 5} - \frac{x^6}{\lfloor 7} \cdots\right)$$

$$= -\left(1 - \frac{x^2}{\lfloor 2} + \frac{x^4}{\lfloor 4} - \frac{x^6}{\lfloor 6} + \cdots\right) + \left(1 - \frac{x^2}{\lfloor 3} + \frac{x^4}{\lfloor 5} - \frac{x^6}{\lfloor 7} + \cdots\right)$$

$$= -\cos x + \frac{1}{x}\left[x - \frac{x^3}{\lfloor 3} + \frac{x^5}{\lfloor 5} - \frac{x^7}{\lfloor 7} + \cdots\right]$$

$$= -\cos x + \frac{1}{x}\sin x.$$

$$\therefore \sqrt{\frac{\pi x}{2}}\, J_{\frac{3}{2}}(x) = \cancel{\cos x + \frac{1}{x}}\ \frac{\sin x}{x} - \cos x$$

$\boxed{\text{proved}}$

Rodrigue's Formula

Rodrigue's formula:- The differential equation $P_n(x) = \frac{1}{2^n \lfloor n} \frac{d^n}{dx^n}(x^2-1)^n$ is called the Rodrigue's Formula.

* we know that,

$$P_n(x) = \frac{1}{2^n \lfloor n} \frac{d^n}{dx^n}(x^2-1)^n$$

Now, when $n=2$.

$$P_2(x) = \frac{1}{2^2 \lfloor 2} \frac{d^2}{dx^2}(x^2-1)^2$$

$$= \frac{1}{8} \frac{d^2}{dx^2}(x^4-2x^2+1)$$

$$= \frac{1}{8} \frac{d}{dx}(4x^3-4x)$$

$$= \frac{1}{8}\left[12x^2-4\right]$$

$$= \frac{1}{8} \times 4\left[3x^2-1\right]$$

$$= \frac{1}{2}\left[3x^2-1\right]$$

Again, when $n=1$.

$$P_1(x) = \frac{1}{2^1 \lfloor 1} \frac{d^1}{dx^1} (x^2-1)^1$$

$$= \frac{1}{2} \frac{d}{dx} (x^2-1)$$

$$= \frac{1}{2} \left[2x\right]$$

$$= x \quad \underline{\underline{\text{Ans}}}$$

Again, when, $n=0$.

$$P_0(x) = \frac{1}{2^0 \lfloor 0} \frac{d^0}{dx^0} (x^2-1)^0$$

$$= 1$$

$\underline{\underline{\text{Ans}}}$

* Given that,

$$\int_{-1}^{1} [P_n(x)]\, dx = 0 \quad \text{when } n \neq 0$$
$$= 2 \quad \text{when } n = 0.$$

We,

$$P_n(x) = \frac{1}{2^n \lfloor n} \frac{d^n}{dx^n} (x^2-1)^n$$

Now Integrating both sides w.r.t. x from -1 to 1 we get,

$$\int_{-1}^{1}[P_n(x)]dx = \int_{-1}^{1}\left[\frac{1}{2^n \lfloor n} \cdot \frac{d^n}{dx^n}(x^2-1)^n\right]dx \quad \text{--- (1)}$$

when $n \neq 0$, then from (1)

$$\int_{-1}^{1}[P_n(x)]dx = 0.$$

when $n = 0$, the from (1)

$$\int_{-1}^{1}[P_n(x)]dx = \int_{-1}^{1}\left[\frac{1}{2^0 \lfloor 0} \cdot \frac{d^0}{dx^0}(x^2-1)^0\right]dx$$

$$= \int_{-1}^{1} dx = [x]_{-1}^{1} = 1+1 = 2$$

$$\therefore \int_{-1}^{1}[P_n(x)]dx = 0 \quad \text{when } n \neq 0$$

$$= 2 \quad \text{when } n = 0.$$

[showed]

* Lagrange's method.

Given that,

$$yzp + xzq = xy$$

where, $P = yz$, $Q = xz$ and $R = xy$

Auxilary equations are,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

ie $\frac{dx}{yz} = \frac{dy}{xz} = \frac{dz}{xy}$

from first two,

$$\frac{dx}{yz} = \frac{dy}{xz}$$

$$\Rightarrow \frac{dx}{y} = \frac{dy}{x}$$

integrating,

$$xdx = ydy.$$

$$\Rightarrow \frac{x^2}{2} = \frac{y^2}{2} + c_1$$

$$\Rightarrow x^2 - y^2 = 2c_1 = u$$

Similary from last two

$$ydy = zdz$$

$$\Rightarrow \frac{y^2}{2} = \frac{z^2}{2} + c_2$$

$$\therefore y^2 - z^2 = 2c_1 = v$$

Hence the solution,

is $\phi(x^2-y^2, y^2-z^2) = 0$ Ans

* Given that,

$$(x^2-yz)p + (y^2-zx)q = z^2-xy$$

A.E are, $\frac{dx}{x^2-yz} = \frac{dy}{y^2-zx} = \frac{dz}{z^2-xy}$

this gives,

$$\frac{dx-dy}{x^2-yz-y^2-zx} = \frac{dy-dz}{y^2-zx-z^2+xy} = \frac{dz-dx}{z^2-xy-x^2+yz}$$

$$\Rightarrow \frac{dx-dy}{(x+y)(x-y)+z(x-y)} = \frac{dy-dz}{(y+z)(y-z)+x(y-z)} = \frac{dz-dx}{(z+x)(z-x)+y(z-x)}$$

$$\Rightarrow \frac{dx-dy}{(x-y)(x+y+z)} = \frac{dy-dz}{(y-z)(x+y+z)} = \frac{dz-dx}{(z-x)(x+y+z)}$$

$$\Rightarrow \frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} = \frac{dz-dx}{z-x}.$$

Now from first two,

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$$

$$\Rightarrow \frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z}$$

$$\Rightarrow \log(x-y) = \log(y-z) + \log c_1$$

$$\Rightarrow \frac{x-y}{y-z} = c_1 = u.$$

and last two,

$$\frac{d(y-z)}{y-z} = \frac{d(z-x)}{z-x}$$

$$\Rightarrow \log(y-z) = \log(z-x) + \log c_2$$

$$\Rightarrow \frac{y-z}{z-x} = c_2 = v$$

The complitions is

$$\phi\left(\frac{x-y}{y-z}, \frac{y-z}{z-x}\right) = 0.$$

Ans

* Higher order linear partial Differential

Given that,

$$(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$$

A.E is

$$m^2 - 6m + 9 = 0$$

$$\Rightarrow (m-3)^2 = 0$$

$$\therefore m = 3, 3$$

$$\therefore C.F = \phi_1(y+3x) + x\phi_2(y+3x)$$

Now,

$$P.I = \frac{1}{(D^2 - 6DD' + 9D'^2)}(12x^2 + 36xy)$$

$$= \frac{1}{(D-3D')^2}(12x^2 + 36xy)$$

$$= \frac{1}{D^2\left(1 - \frac{3D'}{D}\right)^2}(12x^2 + 36xy)$$

$$= \frac{1}{D^2}\left(1 - \frac{3D'}{D}\right)^2 (12x^2 + 36xy)$$

$$= \frac{1}{D^2}\left(1 + \frac{6D'}{D} + \cdots\right)(12x^2 + 36xy)$$

$$= \frac{1}{D^2}(12x^2 + 36xy) + \frac{6D'}{D^3}(12x^2 + 36xy)$$

$$= x^4 + 6x^3y + \frac{6}{D^3}36x$$

$= x^4 + 6x^3y + 9x^4$

$= 16x^4 + 6x^3y.$

Therefore complete solⁿ is

$$z = \phi_1(y+3x) + x\phi_2(y+3x) + 16x^4 + 6x^3y.$$

* Application

~~let~~ Solution:-

let y be the number of Bacteria at the present time t.

then, the function will be

$$y = \phi(t) \text{ ——— } (1)$$

Now, according to the question,

$$\frac{dy}{dt} \propto y$$

$$\Rightarrow \frac{dy}{dt} = ky$$

$$\Rightarrow \frac{dy}{dt} - ky = 0 \text{ ——— } (2)$$

which is the linear differential equation and satisfies eqⁿ (1)

* Now, Integrating factor,

$$I.F = e^{\int -k\,dt}$$

$$= e^{-kdt}$$

Multiplying eq^n ② by I.F

$$e^{-kdt}\frac{dy}{dt} - kye^{-kdt} = 0$$

or, $\frac{d}{dt}\left(e^{-kt} \cdot y\right) = 0$

Now, Integrating

$$e^{-kt} \cdot y = c$$

$$y = ce^{kt} \text{ ——— ③}$$

Applying initial conditions,

when $t=0$, then $y = y_0$

putting in eq^n ③ we get,

$$c = y_0$$

Hence, eq^n ③ becomes

$$y = y_0 e^{kt} \text{ ——— ④}$$

Now, applying the additional condition to find the constant proportionality k

when, $t = 1\,hr$ then $y = 2y_0$

putting in eq^n (4), we get,

$$2y_0 = y_0 e^{kt}$$

$$e^k = 2$$

$$k = \log 2$$

Hence (4) becomes,

$$y = y_0 e^{t \log 2}$$

Now, for final requirement substituting

$y = 3y_0$

$$3y_0 = y_0 e^{t \log 2}$$

$$3 = e^{t \log 2}$$

$$t \log 2 = \log 3$$

$$t = \frac{\log 3}{\log 2} = 1.58\ hr$$

Hence, the number of Bacteria will be triple in 1.58 hrs.

*Solution :-

let T be the temperature of the body at the time t.

Now, according to Newton's law of cooling, the differential equation governing the present situation is

$$-\frac{dT}{dt} \propto (T - T_0) \text{ ——— } (1)$$

T_0 denotes the room temperature $[\because T_0 = 70°F]$

Now, eqn (1) can be written

$$\frac{dT}{dt} = k(T - T_0), \text{ where } k \text{ is constant}$$

or, $\frac{dT}{T - T_0} = k.dt.$ [separating the variable]

Now, Integrating,

$$\log(T - T_0) = kt + A.$$

or, $T - T_0 = e^{kt+A}$

or, $T - T_0 = e^{kA} . e^{A}$

$$T = T_0 + Ce^{kt} \text{ ——— } (2)$$

Applying initial condition,
when temperature measured first
$t=0, \; T=94.6'F$
putting in eq^n (2) we get,

$$94.6=70+ce^{k\cdot 0}$$

$$c=24.6$$

$$\therefore c=24.6$$

Applying addition condition,
$t=1h, \; T=93.4^{\circ}F$
putting is eq^n (2), we get,

$$93.4=70+24.6e^{k\cdot 1}$$

$$e^k=\frac{23.4}{24.6}$$

$$k=\log\frac{23.4}{24.6}$$

$$=-0.05.$$

Hence, the eq^n (1) becomes,

$$T=70+24.6e^{-0.05t} \quad \text{(3)}$$

For, finding the time period since the death substituting; T = 98.6°F in eqn (3)

$$98.6 = 70 + 24.6e^{-0.05t}$$

or, $e^{-0.05t} = \frac{28.6}{24.6}$

or, $t = \frac{0.15}{-0.05}$

or, $t = -3$

$\therefore$ t = 3 hr before,

Therefore, the estimated time of death is 11.30 − 3.00 = 8.30 p.m (approximate)